Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/02/2025
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	Phase 2: ZBOT-2 Thermometry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	05/01/2021	**End Date:**	06/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Pearlman, Howard G.	**Contact Phone:**	216-433-4000
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2026 per H. Pearlman/GRC (Ed., 6/10/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, David A Ph.D. (Michigan State University)		
Grant/Contract No.:	80NSSC21M0128		
Performance Goal No.:			
Performance Goal Text:			
	[Note: This project is part of a program known as “ZBOT-NC”. Per NASA Glenn Research Center, the "NC" stands for "Non-Condensable" since the PI team is adding a non-condensable gas to the test tank to see how the presence of the gas affects the rate at evaporation and condensation of the liquid at the liquid-vapor interface. (Ed., 6/10/25).] In-orbit or on-surface cryogenic storage of propellant and life support consumables is essential to all stages of NASA’s Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment on the International Space Station (ISS) is designed to examine the ZBO concept and, along with computational fluid dynamics (CFD) modeling efforts, provide information on the fundamental physics of the		

Task Description:	underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The optical diagnostics for whole-field thermometry that are being developed in the current proposal is crucial to providing scientific data and measurements for validation and further refinement of the two-phase CFD models developed by the ZBOT microgravity investigation. The ZBOT project directly aligns with NASA exploration needs for Cryogenic Fluid Management of future propellant and life support storage tanks, and the ZBO tank pressure control is an important technology achievement that enables human missions to Moon, Mars, and other planets. Under NASA support we have successfully demonstrated the feasibility of a novel non-intrusive optical method for whole-field temperature measurement suitable for the ZBOT microgravity experiment. The technique of Quantum Dot Thermometry (QDT), specifically developed for the planned ZBOT experiments on the ISS, relies on laser induced fluorescence (LIF) thermometry based on quantum dots (QD) that are solubilized/dispersed in the working fluid of the ZBOT experiment. We have quantified the influence of the various interconnected parameters, and the tradeoffs among them, that influence the ability to make accurate temperature measurements, and we have shown that the science requirements are expected to be met by the judicious selection of detection strategy and data processing methods. The objectives of the proposed research are to capitalize on our progress to date and further develop the technique with a focus on the tradeoffs connected to the actual engineering constraints in implementation and design optimization of experiments on the ISS. A core element of this effort will be the close collaboration with NASA engineering and hardware design experts, and providing our scientific know-how in fluid flow optical diagnostics and data processing/analysis, as we transition the measurement technique to actual implementation for the ZBOT experiments on the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ZBOT project directly aligns with NASA exploration needs for Cryogenic Fluid Management of future propellant and life support storage tanks, and the ZBO tank pressure control is an important technology achievement that enables human missions to Moon, Mars, and other planets. Development of Quantum Dot Thermometry enables the measurements needed for obtaining scientific data for validation and refinement of the two-phase two-component CFD models being developed as part of the ZBOT project.
Task Progress:	Over the last several years under NASA support, we have developed the novel optical method of Quantum Dot Thermometry (QDT) for non-intrusive whole-field mapping of the temperature field in the Zero Boil-Off Tank (ZBOT) microgravity experiment. The technique relies on ratiometric laser induced fluorescence (LIF) thermometry based on quantum dots (QD) that are solubilized/dispersed in the working fluid of the ZBOT experiment, Perfluoro-n-Pentane (PnP). The objectives of the current research effort are to capitalize on our progress to date and further develop the QDT technique for the ZBOT-NC experiment with a focus on the tradeoffs connected to the actual engineering constraints in implementation and optimization of experiments aboard the International Space Station (ISS). A core element of this effort is the close collaboration between our scientific know-how in fluid flow optical diagnostics, data processing/analysis, and the NASA engineering and hardware design experts, as we transition the measurement technique to actual implementation for the ZBOT-NC experiments aboard the ISS. In the first year of this effort, we reported on the identification of QDs that are suitable for QDT and can be solubilized/dispersed in PnP. Characterization of the QDs continued in the second year and led to determination of the key attributes of the QDs that make for ideal candidates for QDT, and more importantly, for the particular camera selected for ZBOT-NC experiments. While a color camera with a Bayer filter is not the preferred implementation of QDT for achieving the highest temperature sensitivity, factors connected to implementation constraints for ISS experiments have led to it being the optimum choice for ZBOT-NC. This puts additional emphasis on optimizing the performance of the QD that works best with the particular Bayer filter color camera. Finalization/optimization of the QD for flight is ongoing. During the second year of this effort, we conducted two separate flight tank tests at ZIN Technologies, the first in April of 2022 and the second in September of 2022. The initial testing identified several aspects of the initial ZBOT-NC optical design that were detrimental to QDT performance. We tested attributes of the system in conditions as similar as possible to their flight condition, such as the laser illumination uniformity and lens vignetting, and established the overall QDT performance curves for the entire system, which determine the expected temperature uncertainty of the measurements. The second round of testing showed notable improvements in the design but also highlighted that further optimization of the components was still needed. In addition to the flight tank testing, a self-pressurization experiment and sub-cooled jet experiment were conducted within the ZBOT fluid breadboard at ZIN. These were representative of the type of experiments being planned aboard the ISS. These preliminary measurements demonstrated promising results even with the less-than-ideal arrangement of the illumination and imaging hardware. More importantly, these collaborative tests demonstrated the initial performance of QDT with the flight hardware in a fluid system similar to the flight hardware. The results of the breadboard tests have been shared with the entire ZBOT-NC team. During the third year of this effort, progress on three distinct areas were made. First, the final batches of QDs were synthesized and characterized, the outcome of which was the final QD selection for flight operations. Second, we made progress in the development of QDT processing approaches that utilize detailed camera response calibrations to recover data in situations that were previously deemed unreliable. Third, several 1g test cases from the ZBOT-NC test matrix were performed at ZIN Technologies within their fluid breadboard. Their breadboard system received several improvements that more closely represented the flight system and was instrumented such that the actual test boundary conditions were sufficiently characterized for Computational Fluid Dynamics (CFD) to simulate the actual test conditions. We focused our initial efforts on the self-pressurization test point with 0.8-Watt strip heaters. Quantitative comparisons between the QDT temperature fields and the CFD simulations were presented at the Annual Meeting of the American Society for Gravitational and Space Research (ASGSR 2023). Overall, the comparisons between the experiments and simulations were excellent. The side-by-side videos of the temperature field evolution over time were complemented with temperature profiles at selected times that highlighted the agreement between the QDT measurements and CFD simulations in the temperature gradients within the tank over the duration of the experiment. These were the first quantitative comparisons made between QDT and CFD for the ZBOT-NC project. The lack of suitable reference conditions during the 1g experiments presented several challenges that were overcome in

	order to produce the QDT data in sufficient fidelity to quantitatively compare to the CFD simulations. Our scientific investigation into the root cause of these issues is ongoing, but the initial guidance on best practices to avoid or mitigate the issues are being provided to NASA/ZIN. The progress made by a summer research intern with successfully applying the camera response calibration yielded additional insight into the root cause of observed issues and is being investigated for possibly providing an alternative pathway to address some of these challenges. During the fourth year of this effort, we continued working with the CFD team on the analysis and comparison of the data from the jet mixing portion of the ground tests. Overall, the Science Team was pleased with these initial comparisons with QDT data from this ground test. The cooperation and iteration between the Michigan State University (MSU) and CFD teams demonstrated the importance of whole field QDT thermometry in guiding the CFD model validation, and improvement. These results were presented at the ASGSR 2024 conference. We continued our work with the engineering team to support their preparations of the flight hardware and flight operations. We made our own preparations by optimizing our QDT processing software to improve our ability to provide timely feedback during flight operations. We have also started processing the Mission Sequence Test QDT data, which were obtained in February of 2025. We have started to identify and provide input on any deficiencies in procedures that need to be improved upon for flight operations.
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Abstracts for Journals and Proceedings	Olson D, Koochesfahani M. "Optimization of optical filters for ratiometric laser induced fluorescence thermometry using a standard color camera." 77th Annual Meeting of the Division of Fluid Dynamics, November 24-26, 2024, Salt Lake City, Utah. Bulletin of the American Physical Society. 2024; Abstract: L23.00011. https://meetings.aps.org/Meeting/DFD24/Session/L23.11 , Nov-2024
Abstracts for Journals and Proceedings	Hylton S, Olson D, Kahwaji Janho M, Winter R, Kartuzova O, Koochesfahani M, Kassemi M. "Ground-based ZBOT-NC self-pressurization & jet mixing test results and CFD model validation against quantum dot thermometry (QDT) measurements." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024